Proximal Gradient Alg: Overview

Composite Optimization: $\min: f(x) = \underset{\text{smooth}}{g(x)} + \underline{h(x)}$

|| Prox Operator for a fn h:

$$\text{prox}_{\eta h} : \mathbb{R}^n \to \mathbb{R}^n$$

$$\text{prox}_{\eta h}(x) \triangleq \arg\min_u : h(u) + \frac{1}{2\eta}\|x-u\|_2^2$$

Prox Grad Alg:

$$x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))$$

The Prox Operator: Example 1

$$h(x) = \|x\|_1 = \sum |x_i|$$

$$\text{prox}_{\eta h}(x) = \text{soft-thresholding operator}$$

$$= z \quad : \quad z_i = \begin{cases} x_i - \eta & x_i > \eta \\ 0 & -\eta \le x_i \le \eta \\ x_i + \eta & x_i < -\eta \end{cases}$$

The Prox Operator: Example 2

$$h(x) = \frac{1}{2} x^T Q x + q^T x + q_0$$

$$\text{prox}_{\eta h}(x) \triangleq \underset{u}{\text{argmin}}: \ \frac{1}{2} u^T Q u + q^T u + q_0 + \frac{1}{2\eta} \| u - x \|_2^2$$

$$\nabla [\quad] = 0 \quad :$$

$$Qu + q + \frac{1}{\eta}(u - x) = 0$$

$$\text{prox}_{\eta h}(x) = (I + \eta Q)^{-1}(x - \eta q).$$

The Prox Operator: Example 3

$$h(x) = I_{\mathcal{X}}(x) = \begin{cases} 0 & x \in \mathcal{X} \\ +\infty & x \notin \mathcal{X} \end{cases}$$

$$\begin{array}{l} \min: g(x) \\ \text{st}: x \in \mathcal{X} \end{array} \iff \min: g(x) + h(x)$$

$$\text{prox}_{\eta h}(x) = \text{argmin}: I_{\mathcal{X}}(u) + \frac{1}{2\eta}\|u - x\|_2^2$$

$$= \begin{array}{l} \text{argmin}: \frac{1}{2\eta}\|u - x\|_2^2 \\ \text{st}: \quad u \in \mathcal{X} \end{array} = \text{proj}_{\mathcal{X}}(x).$$

$h = I_{\mathcal{X}}$: prox = projection.

The Prox Grad Algorithm

min: $g(x) + h(x)$, $g(x)$ smooth

$h(x)$ has "easy" prox fn.

Recall: min: $f(x)$ instead: $x_{t+1} = \text{argmin}: f(x_t) + \nabla f(x_t)^T (x - x_t) + \frac{1}{2\eta} \|x - x_t\|_2^2$

Prox Grad: Same idea – replace g by linear (1st order) approx.

$$x_{t+1} = \underset{x}{\text{argmin}}: \underline{g(x_t)} + \nabla g(x_t)^T (x - x_t) + \frac{1}{2\eta} \|x - x_t\|_2^2 + h(x)$$

$$= \underset{x}{\text{argmin}}: \frac{1}{2\eta} \underbrace{\left(\|x - x_t\|_2^2 + 2\eta \nabla g(x_t)^T (x - x_t) + \underline{\eta^2 \|\nabla g(x_t)\|_2^2} \right)} + \underline{h(x)}$$

$$= \underset{x}{\text{argmin}}: h(x) + \frac{1}{2\eta} \left(\underline{\|x - (x_t - \eta \nabla g(x_t))\|_2^2} \right) \quad //$$

$$= \text{prox}_{\eta h} \left(x_t - \eta \nabla g(x_t) \right).$$

Prox Grad as a Generalization of PGD

$$\min: f(x) \quad \text{st}: x \in \mathcal{X} \quad \Longleftrightarrow \quad \min: \underbrace{f(x)}_{g} + \underbrace{I_{\mathcal{X}}(x)}_{h}$$

$$\text{prox grad}: \quad x_{t+1} = \text{prox}_{\eta h}\left(x_t - \eta \nabla f(x_t)\right)$$

$$= \text{proj}_{\mathcal{X}}\left(x_t - \eta \nabla f(x_t)\right)$$

$$= \text{PGD}.$$

Prox Grad: Summary

If $f = g + h$

g: smooth

h: easy prox.

same step size

same rates of convergence